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Introduction to APIs 45m

In web development, when we talk about APIs, we often refer to an architectural style created by Roy Fielding in 2000 (Fielding, 2000).^{fnref:332a17} This style is called REST (Representational State Transfer) or RESTful architecture.

A REST API (or RESTful API) is a widely used method for connecting the client side and server side to exchange data. It doesn't depend on any specific technology or software but works by using standard HTTP/S methods like GET, POST, PUT, DELETE and so on.

These methods are paired with URIs (Uniform Resource Identifiers) to identify resources or services. It's important to note that REST isn't the same as HTTP – it simply uses HTTP methods as part of its design.

On the client side, you'd write code to make an HTTP/S request using one of these methods to perform an 'operation'. For example, the code could be written in JavaScript running in a browser.

An operation might involve interacting with a service that provides information about objects. Some examples of operations could include:

- **GET /api/objects** – Ask the system to give you **all the objects**.
- **GET /api/objects/:id** – Ask the system to give you **one specific object** (you replace :id with the object's ID number).
- **POST /api/objects** – Tell the system to **add a new object**.
- **PUT /api/objects/:id** – Tell the system to **update the details of a specific object**.
- **DELETE /api/objects/:id** – Tell the system to **remove a specific object**.

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The website restful-api.dev provides examples of live API endpoints (the URLs you use to send requests) that you can try out. You'll notice that most of the responses are returned in JSON format, which is a common way to exchange data. However, REST isn't restricted to using JSON – it can work with other formats too.

Most of the work involved in a REST API happens on the server side, while the client side simply interacts with the API through its interface.

Watch *How to FETCH data from an API using JavaScript* (Bro Code, 2023) (14:16)^{fnref:67a965} to learn about the process of retrieving data from an API using JavaScript.



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REST request

Non-Assessed

Status: Done

Reset ↺

Please modify the code from [Example 7.5 Approach 2](#) to make a REST request to the 'Single object' endpoint, specifically retrieving the item with ID 8 from [REST API - ready to use](#).

☒ I have completed this task

The returned data should be:

```
{"id":"8","name":"Apple Watch Series 8","data":{"Strap Colour":"Elderberry","Case Size":"41mm"}}
```

The key change to the code is the method that needs to be used, which is GET in this example. The URL is the REST endpoint, and the number 8 has been hardcoded into it. However, in most cases, this number would usually be passed into the function that's making the request (much like you clicking on an item on a product page). View the code in [Task 7 REST request](#).

Previous attempts ▼

Dig deeper

You can explore some existing APIs utilised by software like Xero, which provides APIs for its Account microservices. Another example is GitHub, a platform that

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offers its features through a REST API.



Thoughts on this page?

